

A High-Density Low-Leakage and Low-Power Fully Voltage-Stacked SRAM for IoT Application

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Abstract—The general approach to suppress leakage in static random access memory (SRAM) is to use a low voltage (V_L), generated by a low-dropout regulator (LDO), as the cell supply voltage (CVDD) of SRAM array in the standby mode. However, the effectiveness of lowering CVDD is constrained by the area and power overhead introduced by the LDO, as well as the additional latency and power consumption incurred during mode switching. This work presents a fully voltage-stacked (FVS) SRAM that reduces leakage power with internally generated intermediate voltage (V_{MID}) by stacking SRAM arrays. The FVS SRAM consists of a foundry high-density (HD) cell and a pMOS pass-gate (PG) cell having the same metal pattern and size as the HD cell, ensuring compatibility across various process nodes. In addition, the FVS SRAM reduces minimum operating voltage and access energy by the intermediate cell voltage (ICV) assist technique and charge-sharing-based precharge, respectively. The silicon measurement results from a 14-nm FinFET test chip demonstrate that the FVS SRAM achieves a leakage power of 5.34 pW/bit and an access energy of 24.6 fJ/bit at $V_{MIN} = 0.5$ V. Compared to conventional non-stacked SRAM, the FVS SRAM exhibits 29%–59% and 10%–21% reduction in leakage power and access energy across VDD = 0.6–0.8 V.

Index Terms—High density (HD), low leakage, low power, static random access memory (SRAM), voltage stacking (VS).

I. INTRODUCTION

THE rapid growth of Internet-of-Things (IoT) and sensor-driven edge devices has recently gained significant attention, with market forecasts projecting over 34.4 billion connected IoT devices by 2030 [1]. These ultra-low-power devices typically require low-power and low-leakage static random access memory (SRAM) [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16] to support continuous data acquisition, local processing, and wireless communication while maximizing battery lifetime. In addition, high-density (HD) SRAM is increasingly demanded in modern

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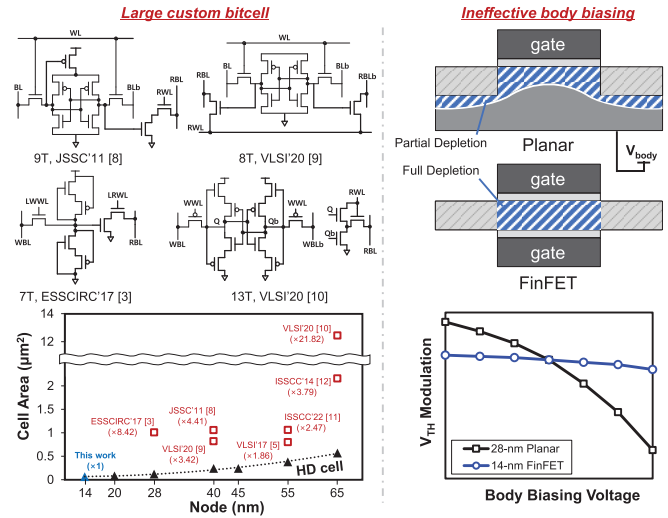


Fig. 1. Limitations of previous leakage suppression techniques.

IoT systems to accommodate the growing need for on-chip memory capacity for buffering excessive sensor data.

Previous studies that focus on leakage suppression use: 1) specialized processes, such as fully depleted silicon-on-insulator (FDSOI) [2], [3], [4], deeply depleted-channel (DDC) [5], [6], and silicon-on-thin-box (SOTB) [7]; 2) large custom cell [3], [8], [9], [10]; and 3) reverse body biasing [3], [4], [6], [9], [11], [12]. However, these approaches face scalability and compatibility issues in advanced nodes (e.g., 7 and 5 nm). Fig. 1 shows the limitations of the previous leakage suppression techniques. A large custom cell can be more than twice the area of a 6T foundry HD cell, significantly reducing the area efficiency. Reverse body biasing modulates the depletion region to increase the threshold voltage (V_{TH}), effectively suppressing leakage current. However, its effect is negligible in recent FinFET technology due to the fully depleted tall fin structure with strong gate controllability [17].

On the other hand, lowering supply voltage (VDD) stands as an effective solution to reduce leakage current with reduced drain-induced barrier lowering (DIBL) effect across various process nodes. Generally, as shown in Fig. 2(a), a conventional non-stacked SRAM employs a low-dropout regulator (LDO) to generate a low voltage (V_L) for the cell supply voltage (CVDD) of the SRAM array. Additionally, peripheral circuits are power-gated during standby mode. However, LDO leakage itself compromises the efficiency of leakage saving and the LDO causes area overhead. Additionally, mode switching

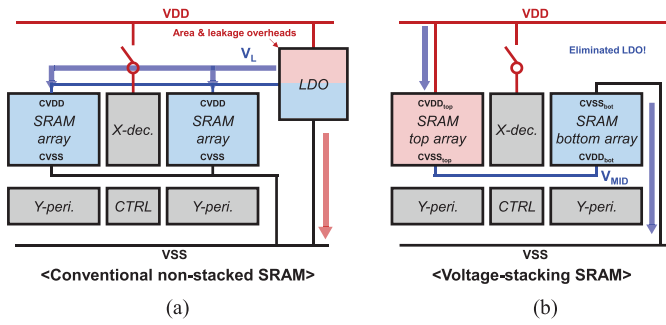


Fig. 2. Methods for reducing leakage power of SRAM. (a) Conventional non-stacked. (b) VS SRAMs.

between normal and standby modes incurs significant latency and power overheads [11], [18] since the LDO regulates the power domains with large capacitance.

Voltage stacking (VS) [19] generates intermediate voltage (V_{MID}) by placing power domains in series, which alleviates the burden on the LDO by recycling charge. This approach has been applied in recent microcontroller [20], high-bandwidth data buses [21], GPUs [22], and SRAM [9]. In SRAM, the arrays can be stacked due to the balanced charge distribution of its symmetric structure, eliminating the need for an LDO [9], as shown in Fig. 2(b). In VS SRAM, the arrays stacked in series are divided into the top and bottom arrays. CVDD of the top array ($CVDD_{top}$) is connected to VDD and cell ground (CVSS) of the bottom array ($CVSS_{bot}$) is connected to VSS. CVSS of the top array ($CVSS_{top}$) is connected to CVDD of the bottom array ($CVDD_{bot}$), forming $V_{MID} = VDD/2$, and thus reducing leakage current. However, the conventional VS SRAM [9] presents several limitations such as a large custom 8T cell, unrecycled bitline (BL) leakage, and the need for array swap process due to inaccessibility of the top array.

This article presents a fully voltage-stacked (FVS) SRAM that stacks the top and bottom arrays in a fully symmetric structure to enable simultaneous access to both arrays. In addition, the proposed design fully recycles the cell leakage by stacking not only power domains but also BLs. The proposed SRAM integrates a foundry HD cell in the bottom array and a custom-designed pMOS pass-gate (PG) cell having identical active and metal layouts in the top array, which ensures excellent compatibility with standard foundry processes. Moreover, access energy is reduced by eliminating the conventional BL precharge through a charge-sharing-based precharge (CS-PCH) scheme. Furthermore, the minimum operating voltage (V_{MIN}) is effectively improved using an intermediate cell voltage (ICV) assist technique without additional yield assist circuits controlling BL, wordline (WL), or cell bias voltages [23], [24], [25], [26], [27], [28], [29], [30], [31].

The remainder of this article is organized as follows. Section II discusses the limitations of previous VS SRAM and the improvements in the proposed FVS SRAM. Section III provides a comprehensive overview of the FVS SRAM architecture. Section IV presents silicon measurement results of 64-kb FVS SRAM macro from a 14-nm FinFET test chip. Finally, Section V summarizes and concludes this article.

II. OVERVIEW OF PREVIOUS AND PROPOSED VOLTAGE-STACKED SRAMs

In VS SRAM, the cell voltage levels vary by array position. In the top array, “0” corresponds to V_{MID} and “1” corresponds to VDD, while in the bottom array, “0” corresponds to VSS and “1” corresponds to V_{MID} . As a result, a different data access method is required to ensure successful read and write operations due to the smaller voltage difference compared to conventional non-stacked SRAM, where “0” and “1” are represented by VSS and VDD, respectively. To address this, the previous VS SRAM [9] employs a custom 8T cell but has several limitations, as shown in Fig. 3(a).

- 1) The large 8T cell ($3.42 \times$ HD cell area) is used to decouple read and write paths, increasing area overhead.
- 2) Leakage reduction is limited since only CVDD and CVSS are stacked, while BLs remain unstacked, resulting in significant PG leakage with $V_{DS} = VDD$.
- 3) The array swap process, which swaps the power domains of top and bottom arrays, is required since the top array is inaccessible. The differential nMOS read buffers cannot be fully turned off when the cell data voltage is at V_{MID} , and clamping current from the unselected row limits the voltage range. This process incurs high energy consumption, long access time, and increased design complexity.

As shown in Fig. 3(b), this work addresses the limitations of previous design as follows.

- 1) The proposed FVS SRAM is composed of a compact foundry HD cell in the bottom array and a custom PMOS PG cell in the top array, both having the same metal pattern and cell size as the foundry HD cell, with all transistors implemented using a single-fin configuration.
- 2) All leakage paths of cells, including BLs, are stacked to enable full recycling of leakage current.
- 3) The use of the pMOS PG cell in the top array ensures robust read stability, thereby enabling direct access and eliminating the need for the inefficient array swap process in [9].

Moreover, the proposed FVS SRAM provides the following additional advantages.

- 1) The symmetric read and write operations of the top and bottom arrays occur simultaneously, enabling CS-PCH to reduce access energy.
- 2) The ICV assist technique enhances read and write margins by controlling cell voltage restoration (CVR) level, resulting in a lower V_{MIN} compared to conventional non-stacked SRAM without assist.
- 3) The proposed FVS SRAM supports bit-interleaving, which improves area efficiency and provides soft error protection [32], [33] by satisfying read and write margins simultaneously. A MUX4 is employed in the 64-kb test macro.

III. PROPOSED FVS SRAM

A. Macro Architecture of FVS SRAMs

Fig. 4 shows the macro architecture, schematics of the top and bottom cells, precharger (PC), and cell voltage controller

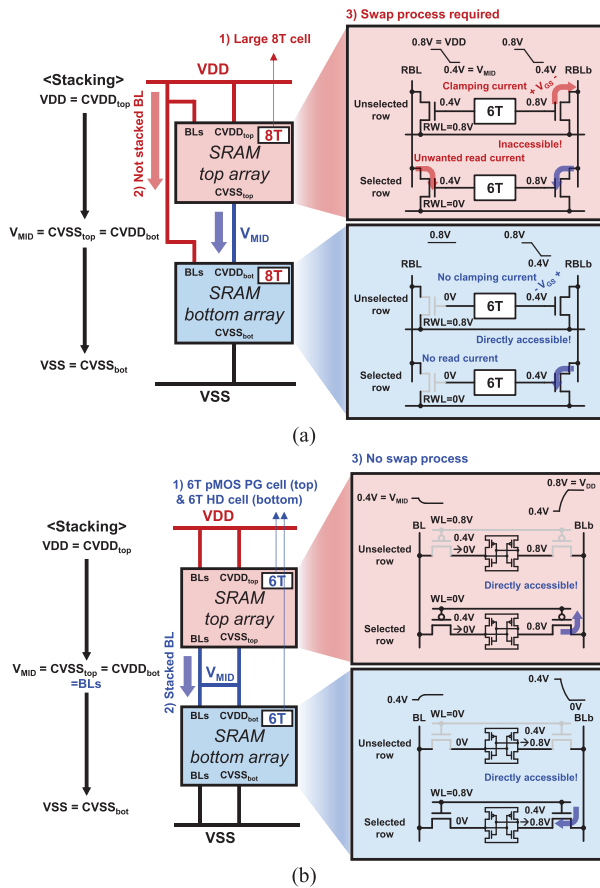


Fig. 3. (a) Previous VS SRAM [10]. (b) Proposed FVS SRAM.

(CVC) in the proposed FVS SRAM. The 128×128 top and bottom arrays are arranged in a butterfly structure, with the top array placed to the left and the bottom array placed to the right of the row decoder. As a result, WL is simultaneously applied to both arrays without additional access energy overhead. WL = “0” activates the pMOS PG in the top array and WL = “1” activates the nMOS PG in the bottom array. The layout of pMOS PG cell used in the top array swaps N-well and P-well positions while maintaining the metal and active patterns of the foundry HD cell, as shown in Fig. 5. This approach can be easily applied across various process nodes. Additionally, the use of fully symmetric 6T cells for both the top and bottom arrays ensures that V_{MID} stability remains robust against variations in the distribution of “0” and “1” data. In contrast, the use of asymmetric cells can cause the leakage current to vary depending on the stored data, which may result in V_{MID} fluctuations influenced by the data pattern.

A dummy row is inserted every eight rows to segment the vertically routed $CVSS_{top}$ and $CVDD_{bot}$ power lines, reducing CVR time and energy with minimal area overhead. The CVCs placed between the WL decoder within a single cell pitch connect $CVSS_{top}$ and $CVDD_{bot}$ of each sub-array to the V_{MID} node in the hold state. In other words, the CVC enables series connection of power domains when the awake signal (AWK) is “0.” BLs are also stacked through the PC in the Y-peripheral circuits when the precharge signal (PCH) is “1.” Unlike conventional pMOS PCs, it uses transmission gates to

prevent V_{TH} drop, ensuring that BLs charge or discharge to $V_{MID} = VDD/2$ by the charge-sharing operation.

The rest of the peripheral circuits consist of a controller (CTRL), a sense amplifier (SA), a write driver (WD), and latched input-output buffers.

B. Leakage Power Reduction With Voltage-Stacked Structure in Hold State

In the hold state, the bias voltage of top and bottom cells is lowered from VDD to V_{MID} by connecting the power domains and BLs of top and bottom arrays in series with CVCs and PCs. Notably, for the pull-down (PD) transistor in the top cell and the pull-up (PU) transistor in the bottom cell, an inherent reverse body biasing increases V_{TH} and further reduces leakage in planar devices. Fig. 6(a) shows the simulated leakage power of the proposed FVS SRAM with and without stacked BLs. The proposed FVS SRAM without BL stacking achieves a 21%–38% reduction in leakage power compared to conventional non-stacked SRAM by stacking power domains, across various process corners and VDD levels. With BL stacking, leakage power is further reduced by 35%–50%, as V_{DS} in the PG is reduced from VDD to V_{MID} .

In the proposed FVS SRAM, achieving V_{MID} level of $VDD/2$ is crucial for cell data retention and stable operations. Fig. 6(b) shows the V_{MID} level of the FVS SRAM. At the FF and SS corners, the V_{MID} level remains close to $VDD/2$ because the effective resistances from V_{MID} to both VDD and VSS are balanced due to the fully symmetric structure, and both transistor types are skewed in the same direction. However, in the FS and SF corners, the V_{MID} deviation increases up to 8.3%. In the FS corner, a large leakage of nMOS leads to increased leakage current in the bottom array, which uses nMOS PG cell. As a result, the V_{MID} level drops below $VDD/2$. Conversely, in the SF corner, a large leakage of pMOS increases the leakage in the top array, which uses a pMOS PG cell. Consequently, the V_{MID} level rises above $VDD/2$.

Fig. 6(c) shows the 6σ cell data retention voltage (DRV) of top and bottom cells. To simulate the DRV, the cell was modeled using the SRAM operational mismatch (SOMM) corner model [34] under 6σ worst case conditions, and post-layout transient SPICE simulation was performed where the cell bias voltage was dynamically lowered and then restored after $1 \mu s$. The characteristics of DRV also exhibit asymmetry between the top and bottom cells. In the FF and SS corners, both transistor types are skewed in the same direction, resulting in DRV levels similar to those in the TT corner. However, in the FS corner, the increased leakage through the nMOS PG leads to a higher DRV in the bottom array compared to other corners. In contrast, in the SF corner, the increased leakage through the pMOS PG causes a greater DRV increase in the top array compared to other corners.

The variations of V_{MID} and DRV across global process corners deteriorate the retention V_{MIN} . Fig. 7 presents the 6σ retention V_{MIN} of the proposed FVS SRAM, and the retention V_{MIN} can be determined using the following condition: DRV of bottom array $< V_{MID} < VDD - DRV$ of top array. The minimum VDD that satisfies this condition is defined as the retention

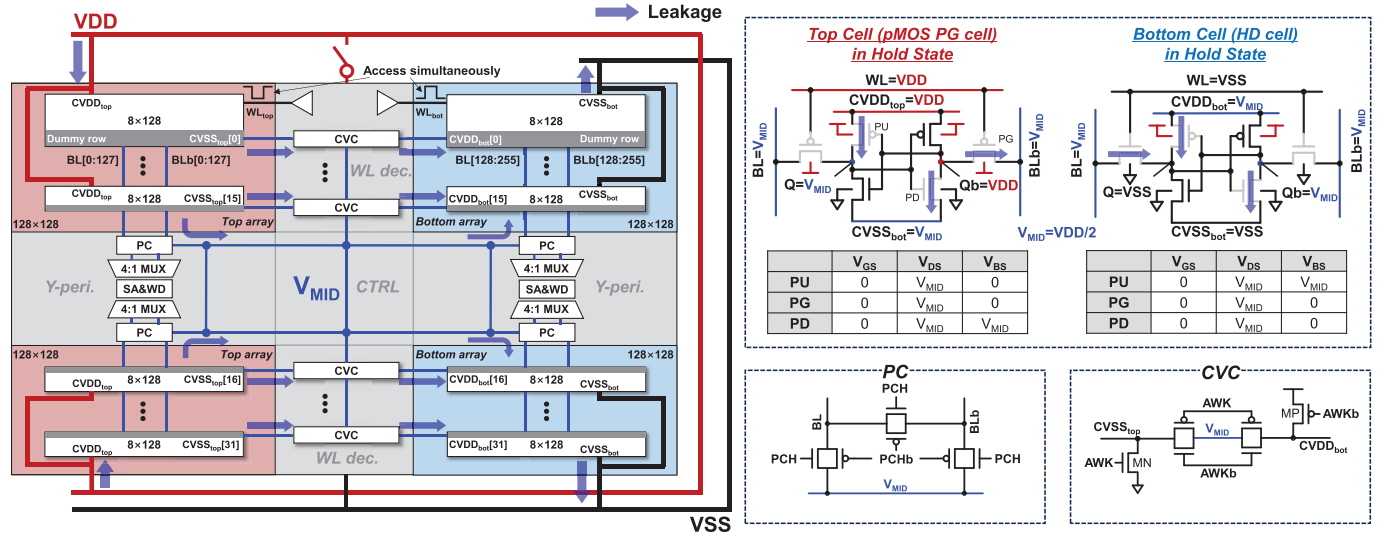


Fig. 4. Proposed macro and schematics of the top and bottom cells, PC, and CVC.

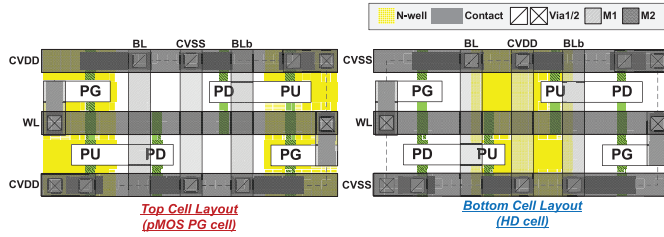


Fig. 5. Layouts of top and bottom cells.

V_{MIN} . In the FS corner, V_{MID} shifts below $VDD/2$, which reduces the voltage applied to the bottom cell. At the same time, DRV of the bottom array increases due to the higher nMOS leakage, resulting in the degradation of retention V_{MIN} due to the opposing effects. In the SF corner, V_{MID} shifts above $VDD/2$, which reduces the voltage applied to the top cell. Meanwhile, DRV of the top array increases due to the higher pMOS leakage, leading to the degradation in retention V_{MIN} . This analysis shows that retention V_{MIN} degradation is most significant under FS and SF corners, due to opposing shifts in both V_{MID} and DRV between the top and bottom arrays. The worst case 6σ retention V_{MIN} is observed to be 0.61 V under the SF corner.

The power supply to peripheral circuits is power-gated by applying pMOS header, except for the CVC and PC. Fig. 8 illustrates the normalized wake-up time and leakage power according to the header size. A header size of 8 was chosen to balance wake-up time and leakage power.

C. Energy-Efficient Read and Write Operations With Charge-Sharing-Based Precharge

Fig. 9(a) shows the waveforms of the write “1” and read “0” operations in the proposed FVS SRAM, performed across three consecutive phases: Ph-1) CVR phase to achieve sufficient operational yield; Ph-2) WL activation phase to access cells; and Ph-3) CS-PCH phase to stack $CVSS_{top}$, $CVDD_{bot}$, and BLs, restoring their voltage levels to V_{MID} .

Ph-1) CVR Phase: In the hold state, the bias voltage for all cells is suppressed to $V_{MID} = VDD/2$ by stacking arrays, which effectively reduces the leakage power. However, at this low voltage level, 6T cells cannot ensure sufficient read and write yields. To address this, the proposed FVS SRAM performs the CVR prior to WL activation. When the AWK signal is asserted, $CVSS_{top}$ and $CVDD_{bot}$ for the selected sub-arrays start the voltage restoration process from V_{MID} toward VSS and VDD, respectively. This CVR process is controlled via MN and MP within the CVC. The CVR operation can affect read stability by introducing fluctuations in $CVSS_{top}$ and $CVDD_{bot}$ due to the limited driving strength of the CVC transistors and the resistance of the metal path, which limits the read current delivered to $CVSS_{top}$ or from $CVDD_{bot}$. However, in the proposed FVS SRAM, the majority of the read current from the top and bottom cells flows directly through VDD ($= CVDD_{top}$) and VSS ($= CVSS_{bot}$), respectively, rather than through weakly restored $CVSS_{top}$ and $CVDD_{bot}$. In the top (bottom) array, the pMOS (nMOS) PG cell charges (discharges) BL from V_{MID} to VDD (VSS). Thus, MN and MP need to provide only a small current to maintain the cell voltage [35], enabling the use of the minimum-sized MN and MP. Due to their small size, the leakage current through MN and MP is negligible. In this work, MN and MP were sized with eight fins considering the CVR speed. In addition, by disconnecting $CVSS_{top}$, $CVDD_{bot}$, and BLs from V_{MID} during read and write operations, dynamic current paths are eliminated, thereby further enhancing V_{MID} voltage stability.

In a conventional non-stacked SRAM array, restoring voltage across unsegmented $CVSS$ or $CVDD$ metals with significant capacitance results in excessive CVR time and energy consumption. To overcome this, the proposed FVS SRAM employs a segmented array architecture by inserting dummy rows, as shown in Fig. 4. These dummy rows segment the array into sub-arrays, thereby reducing the effective metal capacitance of $CVSS_{top}$ and $CVDD_{bot}$. The vertically routed $CVSS_{top}$ and $CVDD_{bot}$ in each column are merged at the dummy row and connected to the CVC.

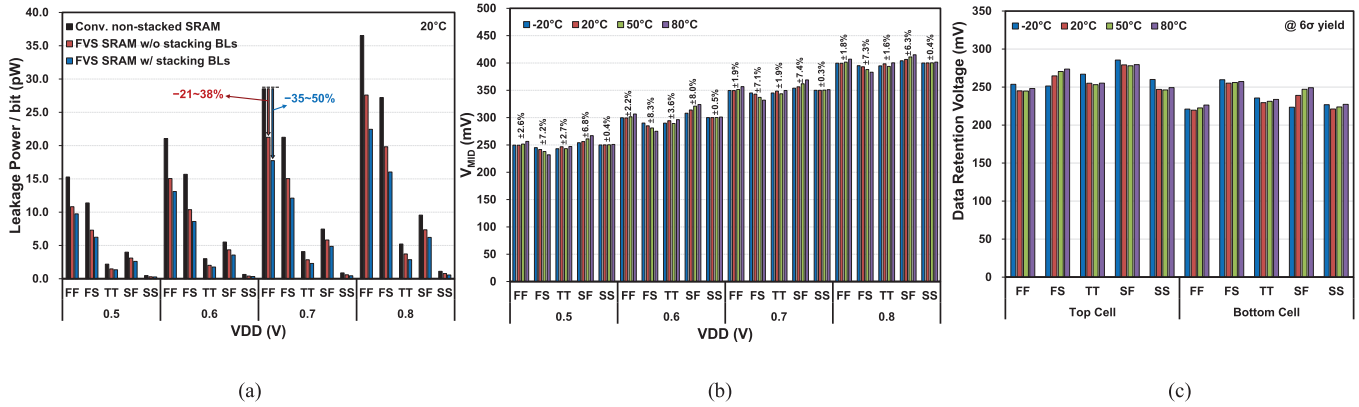


Fig. 6. (a) Leakage power of FVS SRAM with and without stacked BLs. (b) Voltage levels of V_{MID} . (c) DRV of top and bottom cells.

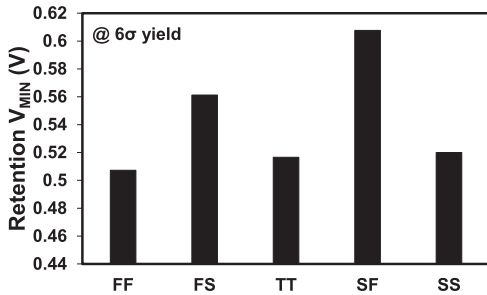


Fig. 7. Retention V_{MIN} of the proposed FVS SRAM.

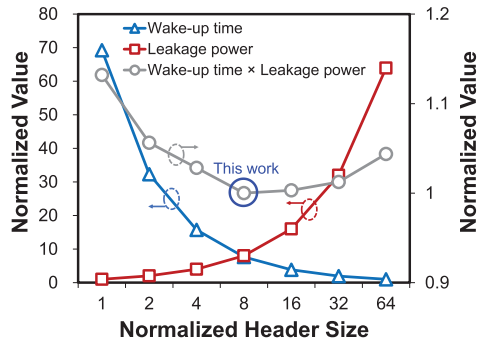


Fig. 8. Wake-up time and leakage power according to header size.

Fig. 10 presents the tradeoff between area increase and CVR time/energy according to SRAM rows per dummy row. As the number of SRAM rows per dummy row increases, the area overhead decreases, but CVR time and energy increase significantly. To achieve an optimal balance, this work adopts eight SRAM rows per one dummy row with an increased area of $1.125\times$. This choice also ensures that the wake-up time from the hold state is hidden within the CVR phase. For the write operation, the CVR phase hides not only the wake-up time but also BL driving time from V_{MID} to VSS or VDD.

Ph-2) WL Activation Phase: After the CVR phase, WL is activated simultaneously for both top and bottom arrays to perform a symmetric read or write operation. During the write operation, the cell data flip according to the BL value. In the top array, the pMOS PG makes the BL at VDD the dominant write path. In contrast, for the bottom array, the BL

at VSS serves as the dominant write path. During the read operation, in the top array, pMOS PG charges one BL from V_{MID} to VDD, while the other remains near V_{MID} due to a V_{TH} drop. In the bottom array, nMOS PG discharges one BL to VSS. Conventional voltage-latch SA (VLISA) could lead to sensing failures due to the sensing dead zone with the BL precharge voltage level of V_{MID} . To address this issue, a VLISA with double switches and transmission gate access transistors (DSTA-VLISA) [36] was employed for stable SA operation across all voltage regions.

Ph-3) CS-PCH Phase: In conventional non-stacked SRAM, significant energy is consumed to precharge the BLs with a large capacitance (C_{BL}) back to VDD. On the other hand, the symmetric read and write operations in the top and bottom arrays of the proposed FVS SRAM effectively reduce access energy with CS-PCH, as shown in Fig. 11.

After the WL activation phase, the voltage levels of $CVSS_{top}$ and $CVDD_{bot}$ are restored to VSS and VDD, respectively. When returning to the hold state, the voltage levels of $CVDD_{bot}$ and $CVSS_{top}$ return to V_{MID} by sharing the charge on $CVDD_{bot}$ with $CVSS_{top}$ through the CVC. Similarly, charge sharing also occurs for the BLs. After the write operation, the voltage levels of the BL and BLb are set to VDD and VSS, respectively, in opposite directions. Thus, BL and BLb can be restored to V_{MID} by charge sharing between themselves. After the read operation, the charge from the BLs above V_{MID} in the top array is shared with the BLs below V_{MID} in the bottom array, as illustrated in Fig. 11.

However, CS-PCH of $CVSS_{top}$, $CVDD_{bot}$, and BLs can occur V_{MID} fluctuations. To mitigate this fluctuation, an $M4$ -to- $M7$ metal-oxide-metal (MOM) capacitor is integrated alongside the WL decoders, avoiding interference with the $M4$ address and power lines. The MOM capacitor occupies an area of $4 \times 63.6 \mu m$ and provides a capacitance of 1 pF. Post-layout transient simulations at $VDD = 0.8$ V verify that the V_{MID} fluctuations to below 6.8 mV, corresponding to less than 1% of VDD.

Fig. 12 shows the normalized energy consumption of read and write operations. The proposed FVS SRAM incurs additional energy consumption compared to conventional non-stacked SRAM due to the following reasons: 1) charging BLs during read and write operations; 2) CVR operation; and 3) extra decoder for AWK signals and the delay chain for

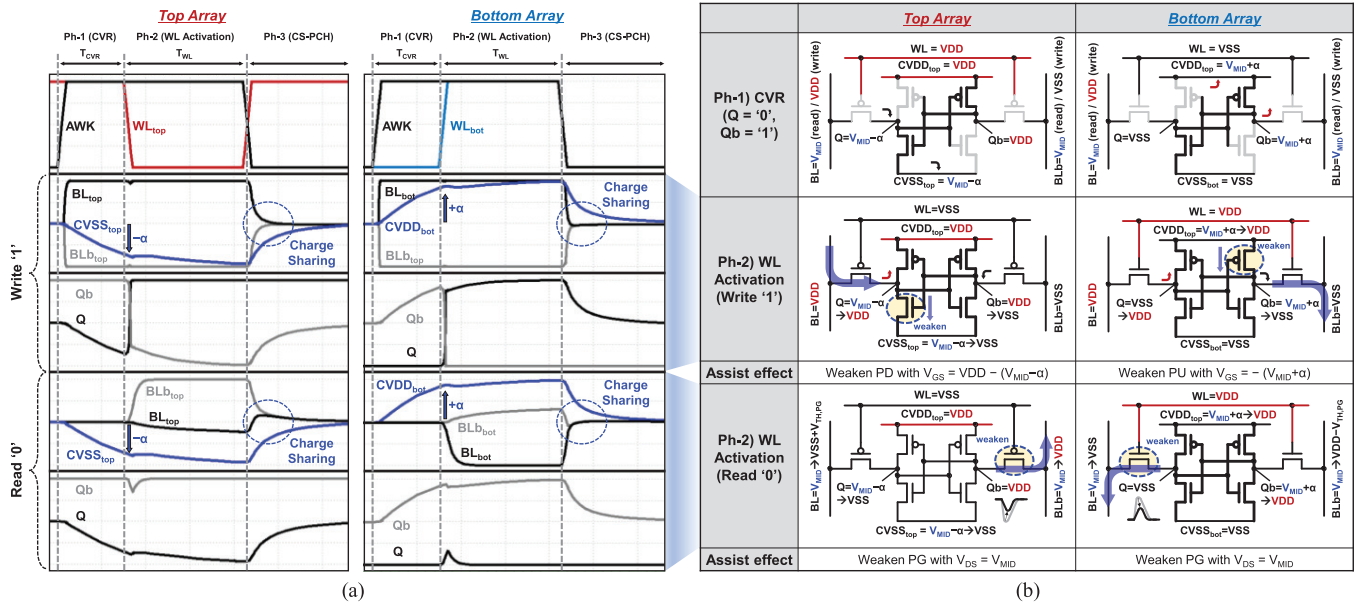


Fig. 9. (a) Waveforms of the write “1” and read “0” operations. (b) ICV assist technique in the proposed FVS SRAM.

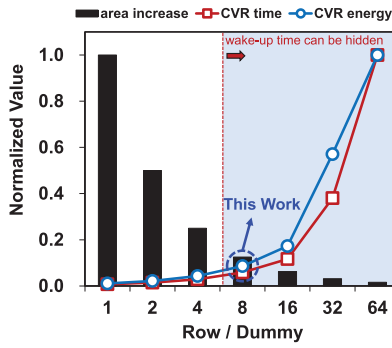


Fig. 10. Tradeoff between area increase and CVR time/energy based on the number of SRAM rows per dummy row.

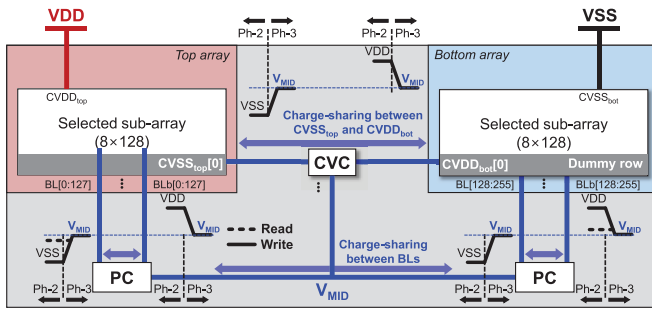


Fig. 11. Charge-sharing-based precharge during ph-3.

CVR phase time (T_{CVR}) generation in the CTRL. However, the elimination of BL precharge energy significantly contributes to reducing overall energy consumption.

In the read operation, the energy consumed by the cell increases because the BL is charged through the pMOS PG in the top array. Additionally, conventional VLSA discharges one output node to VSS while maintaining the other at VDD. However, the DSTA-VLSA amplifies the output node from near V_{MID} to VDD, resulting in higher SA energy consumption.

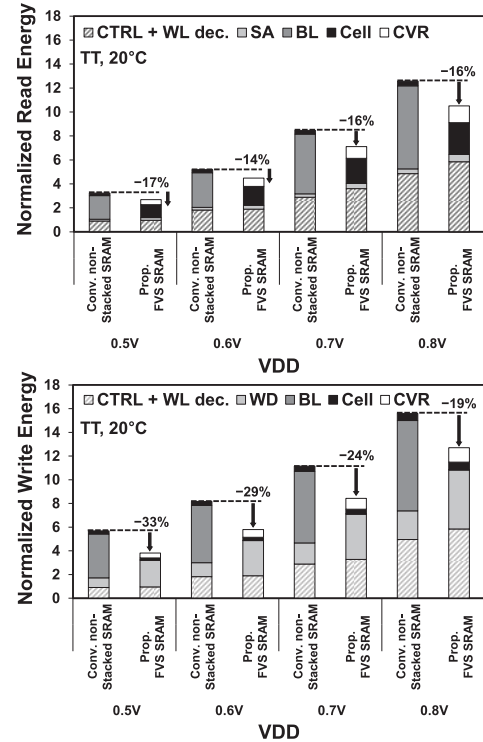


Fig. 12. Energy consumption for read and write operations for different VDD levels.

In the write operation, the proposed FVS SRAM exhibits an increase in WD energy consumption because the WD charges one BL from V_{MID} to VDD. However, compared to the BL precharge energy of $C_{BL} \cdot V_{DD}^2$, the increase in WD energy consumption is insignificant, as it is only $C_{BL} \cdot V_{MID}^2$.

The proposed FVS SRAM achieves a 14%–17% reduction in read energy and a 19%–33% reduction in write energy across different VDD levels by eliminating the BL precharge energy.

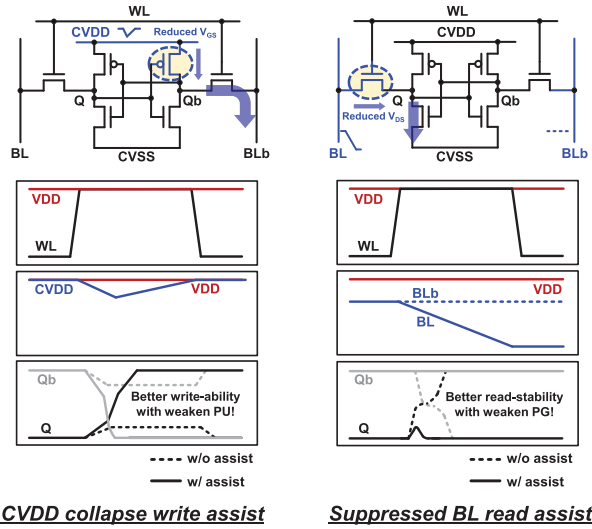


Fig. 13. Conventional write and read assist techniques based on V_{GS} and V_{DS} relaxation.

D. V_{MIN} Improvement With ICV Assist Technique

The proposed ICV assist technique intentionally activates WL when $CVSS_{top}$ and $CVDD_{bot}$ before perfectly restored to enhance read and write yields simultaneously in the bit-interleaved structure. As a result, the ICV assist technique reduces V_{MIN} compared to conventional non-stacked SRAM without assist operating with full cell voltage range of VSS and VDD.

Fig. 13 illustrates the conventional assist techniques that have been widely used to relax V_{GS} and V_{DS} in SRAM operations. First, CVDD collapse [24] reduces the source voltage of the PU during write operation, thereby relaxing its V_{GS} . This relaxation weakens the contention between the cell and the WD through the PG, effectively improving writeability. Second, suppressed BL [37] reduces V_{DS} of the PG during read operation. By reducing the BL precharge voltage, the read current entering the cell is suppressed, thereby enhancing read stability by minimizing voltage bumping at the data “0” node. These conventional techniques serve as the foundation for the proposed FVS SRAM, which achieves V_{MIN} improvement by simply activating the WL before the cell voltages are fully restored and by stacking BLs, without incurring additional area or energy overhead.

The schematic in Fig. 9(b) illustrates the ICV assist effect for write “1” and read “0” scenarios for the top and bottom cells. During the write operation, writeability is enhanced in a manner similar to the CVDD collapse [24] write assist technique by weakening the strength of a cell. In the top array, which employs a pMOS as PG, the write path is established through the BL driven to VDD. During the write operation, the source voltage of the PD is raised above VSS as $CVSS_{top}$ is greater than VSS, resulting in a relaxed V_{GS} . This relaxation weakens the strength of the PD, which facilitates the flipping of the stored “0” to “1” and thereby improves writeability. Similarly, in the bottom array, the weakened PU due to the lowered voltage level of $CVDD_{bot}$ enhances writeability, exhibiting the same effect as CVDD collapse.

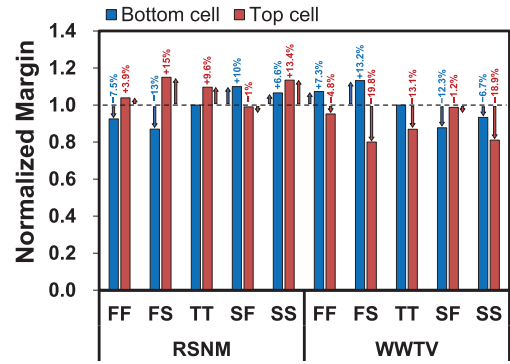


Fig. 14. Normalized read and write margins of top and bottom cells.

During the read operation, read stability is enhanced in a manner similar to the suppressed BL [37] read assist technique by weakening strength of the PG. In the top array, the read path is formed from $Qb = VDD$ to $BLb = V_{MID}$ through the pMOS PG. Thus, the read stability is influenced by the downward voltage bumping at Qb caused by the current flowing through PG. With the relaxed $V_{DS} = V_{MID}$ across the PG during read operation, the resulting mitigation of voltage bumping improves read stability. A similar read assist effect occurs in the bottom array with the weakened nMOS PG. When WL is activated, the drive strength of the PG is weakened due to $V_{DS} = V_{MID}$. This results in a lower magnitude of voltage bumping, providing a read stability assist effect. This read assist effect occurs similarly in the bottom array with the weakened nMOS PG.

The proposed ICV assist technique enhances writeability with a shorter T_{CVR} and improves read stability with a longer T_{CVR} . Thus, optimally balanced T_{CVR} is critical to improving V_{MIN} . For example, in the top array, when $CVSS_{top}$ is assumed to decrease by α after T_{CVR} , V_{GS} of the PD is given by $V_{GS} = VDD - (V_{MID} - \alpha)$. As α increases with T_{CVR} , the strength of PD enhances due to an increase in V_{GS} , resulting in deterioration of writeability. On the other hand, as α increases with T_{CVR} , the strength of PU also enhances with $V_{GS} = -(V_{MID} + \alpha)$, resulting in improvement of read stability. This is because the increased strength of the PU reduces the extent of downward voltage bumping. This dependency of read and write yields on T_{CVR} exhibits the same trend in the bottom array through an identical mechanism.

In the proposed FVS SRAM, operational yield is sensitive to global process corners. Fig. 14 shows the normalized read and write margins of the top and bottom cells, extracted from read static noise margin (RSNM) and WL write trip voltage (WWTV), respectively. At FF and SS corners, where both transistor types are skewed in the same direction, the top cell exhibits higher RSNM but lower WWTV than the bottom cell due to the weaker pMOS PG compared to the nMOS PG. However, RSNM and WWTV shift significantly in FS and SF corners, where nMOS and pMOS are skewed in opposite directions, and the difference between the top and bottom cells becomes largest in the FS corner.

In the FS corner, the bottom cell's RSNM is reduced due to strong nMOS and weak pMOS, which lowers the inverter

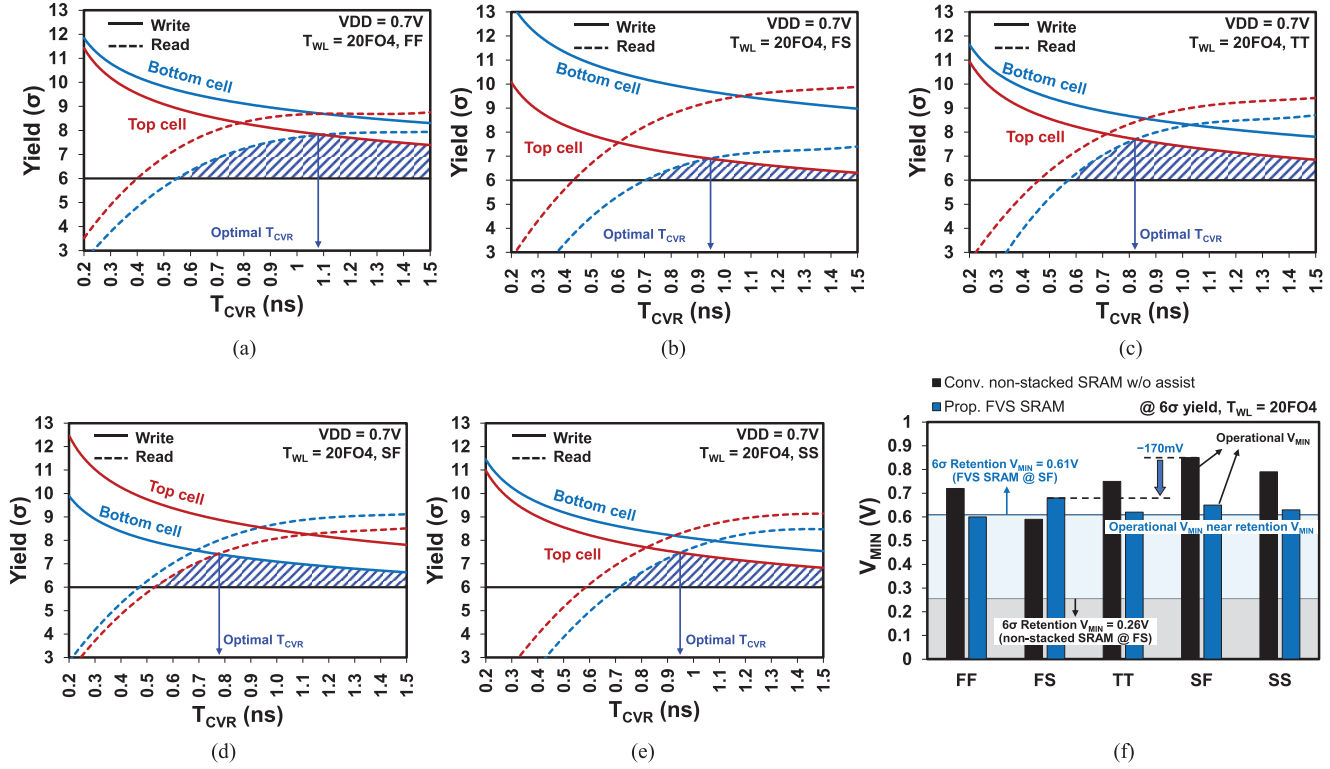


Fig. 15. Simulated read and write yields depending on T_{CVR} at $V_{DD} = 0.7$ V across (a) FF, (b) FS, (c) TT, (d) SF, and (e) SS corners. (f) Operational and retention V_{MIN} of the conventional non-stacked SRAM without assist and the proposed FVS SRAM.

trip voltage (V_{trip}). This makes the cell more susceptible to upward voltage bumping during read operation. Conversely, the top cell's RSNM improves, as the lowered V_{trip} enhances robustness against downward voltage bumping. During write operation in the FS corner, the weak pMOS PU in the bottom cell increases its writeability, while the weak pMOS PG in the top cell reduces its writeability. As a result, the increased difference in read and write yields between top and bottom arrays deteriorates the overall operational yield of the FVS SRAM, in which accesses to both arrays are simultaneously accessed.

Fig. 15(a)–(e) shows the simulated read and write yields depending on T_{CVR} at $V_{DD} = 0.7$ V across global process corners. As T_{CVR} increases, the read yield increases, but the write yield tends to decrease. Thus, the maximum operational yield is obtained at the point where the write yield of the top cell and the read yield of the bottom cell intersect, and this point defines the optimal T_{CVR} . The proposed FVS SRAM adjusts T_{CVR} through the tunable delay chain in the CTRL. At the TT corner, the blue area bounded by the top cell's write yield and the bottom cell's read yield indicates the region where the operational yield exceeds 6σ . In the FS corner, as both top cell's write yield and bottom cell's read yield decrease, the blue area becomes narrow, leading to the degradation in the overall operational yield. In the SF corner, the difference in read and write yields between top and bottom arrays is smaller than in the FS corner due to the inherent yield imbalance of cells.

Fig. 15(f) summarizes the simulated 6σ operational and retention V_{MIN} of the conventional non-stacked SRAM without assist and the proposed FVS SRAM. For the conventional non-stacked SRAM without assist, the operational V_{MIN} is limited by the write yield and exhibits the worst operational V_{MIN} in the SF corner. The proposed FVS SRAM shows the worst operational V_{MIN} in the FS corner, where the difference in read and write yields between top and bottom arrays is largest. The proposed ICV assist technique achieves a 170-mV reduction in operational V_{MIN} compared to the conventional non-stacked SRAM without assist.

The conventional non-stacked SRAM without assist exhibits a large difference between operational and retention V_{MIN} , and thus, standard assist techniques are commonly used to reduce operational V_{MIN} around 200–400 mV [24], [29], [30], [31]. However, in VS SRAMs, standard assist techniques become less effective in terms of area, energy, and design complexity overheads since the increased retention V_{MIN} limits the reduction of operational V_{MIN} , as shown in Fig. 15(f). The proposed ICV assist technique effectively improves operational V_{MIN} closer to the retention V_{MIN} by simply activating the WL during the CVR operation without incurring any additional area or energy overhead.

At high V_{DD} , the reverse read-stability issue caused by increased V_{GS} of the PG exceeding its V_{TH} with $V_{BL} = V_{DD}/2$ can potentially limit the maximum operating voltage (V_{MAX}) of the proposed FVS SRAM. However, a 6σ read yield is ensured at $V_{DD} = 1.4$ V under a low-power 14-nm FinFET

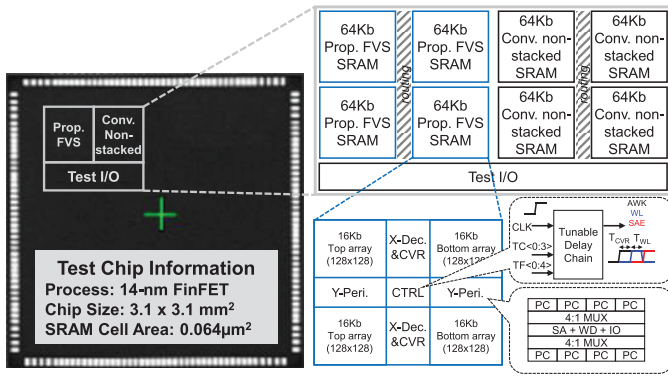


Fig. 16. Test chip photograph and macro configuration.

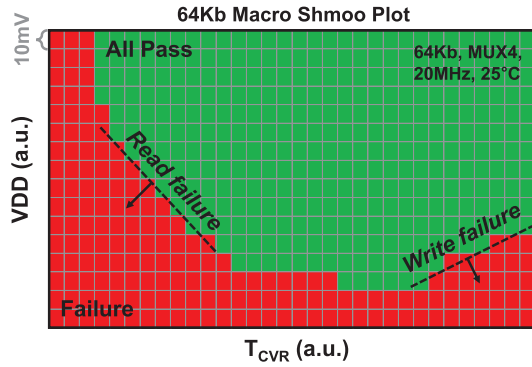


Fig. 17. Measured shmoo plot of VDD versus T_{CVR} for the proposed FVS SRAM.

process, and high VDD operation is less likely to be a critical issue, as energy efficiency is prioritized over performance in IoT applications.

IV. SILICON MEASUREMENTS

Fig. 16 shows the test chip photograph and 64-kb macro configuration. A 64-kb macro consists of two top arrays and two bottom arrays, each 16 kb, arranged in a butterfly structure. The tunable delay chain is integrated into the CTRL to adjust activation timings for AWK, WL, and SA enable (SAE) signals. This enables precise control over T_{CVR} and T_{WL} through coarse and fine delay tuning signals, denoted as $TC<0:3>$ and $TF<0:4>$, respectively. Fig. 17 shows the measured shmoo plot of VDD versus T_{CVR} for the proposed FVS SRAM. The read and write yields of the proposed FVS SRAM vary with T_{CVR} as described in the simulation results presented in Section III-C. The relationship between the operational yield and T_{CVR} was confirmed by measuring the operational VDD according to T_{CVR} . If T_{CVR} is too short, $CVSS_{top}$ and $CVDD_{bot}$ do not reach a sufficient voltage level, causing read failures due to significant voltage bumping when WL is activated. This flips the cell data. Conversely, if T_{CVR} is too long, $CVSS_{top}$ and $CVDD_{bot}$ approach VSS and VDD, respectively, preventing the write assist effect with the voltage collapse. As confirmed by the yield simulation results in Fig. 15, the measured V_{MIN} of the proposed FVS SRAM is determined near the optimal T_{CVR} value, where both the read-stability yield and writeability yield are sufficiently high.

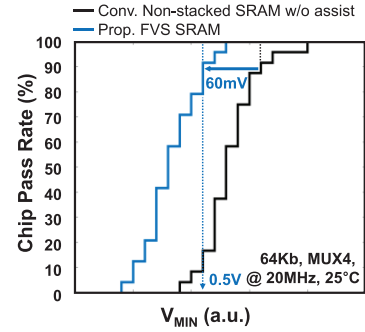


Fig. 18. Measured chip pass rate.

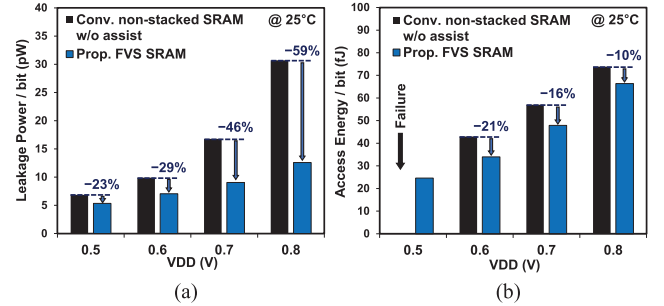


Fig. 19. Measured (a) leakage power and (b) access energy for different VDD levels.

Fig. 18 shows the measured chip pass rate. At a 90% chip pass rate, the proposed FVS SRAM achieves V_{MIN} of 0.5 V, which is 60 mV lower than that of conventional non-stacked SRAM, attributed to the ICV assist technique. The 60-mV improvement is smaller than the simulated 170-mV improvement, as the 64-kb test macro reflects only $\sim 4.3\sigma$ yield and was tested solely with TT corner chips, without accounting for global process corners. Fig. 19 shows the measured leakage power and access energy. The access energy is defined as the average of the read and write energy. The access energy of conventional non-stacked SRAM without assist at VDD = 0.5 V is not plotted because it fails to achieve a 90% chip pass rate at this voltage. At VDD = 0.5 V, the proposed FVS SRAM consumes 5.34 pW/bit of leakage power and 24.6 fJ/bit of access energy. Across VDD = 0.5–0.8 V, the proposed FVS SRAM reduces leakage power by 23%–59% compared to conventional non-stacked SRAM without assist, primarily due to the stacking effect. Additionally, the proposed FVS SRAM achieves a reduction in access energy by 10%–21% across VDD = 0.6–0.8 V, attributed to the CS-PCH following the read and write operations.

Table I provides a comparison of this work with previously reported low-leakage and low-power SRAMs. Previous works have employed various techniques to reduce leakage power, including using an LDO to supply V_L to cells, utilizing specialized processes, or applying reverse body biasing. However, the LDO introduces leakage that diminishes the efficiency of leakage reduction and adds area overhead. The previous SRAMs that rely on leakage-specialized processes may struggle to maintain their reported leakage characteristics when implemented in general process nodes. The reverse body biasing becomes ineffective in advanced FinFET technologies

TABLE I
COMPARISON OF LOW-LEAKAGE AND LOW-POWER SRAMS

	Fukuda, ISSCC'14 [12]	Gupta, VLSI'20 [10]	Dong, VLSI'17 [5]	Pons, CICC'19 [6]	Zhang, ISSCC'22 [11]	Teman, JSSC'11 [8]	Wang, VLSI'20 [9]	Tachibana, ISSCC'13 [13]	Ranica, VLSI'16 [2]	Maeda, VLSI'12 [14]	Frustaci, ISSCC'14 [15]	Haine, ESSCIRC'17 [3]	Joshi, VLSI'17 [4]	Fujiwara, VLSI'13 [16]	This work
Process Node	65-nm	65-nm	55-nm	55-nm	55-nm	40-nm	40-nm	28-nm	28-nm	28-nm	28-nm	28-nm	22-nm	20-nm	14-nm
Cell	6T	13T	6T	6T	6T	9T	8T	6T	6T	6T	6T	7T	6T	6T	6T
LDO Required?	Yes	No	No	Yes	Yes	No	No	Yes	No	Yes	Yes	Yes	Yes	Yes	No
Single-Mode?	No	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	No	No	No	Yes	No	Yes
Process Independent?	No (XXL)	Yes	No (DDC)	No (DDC)	No (body biasing)	Yes	No (body biasing)	Yes	No (FDSOI)	Yes	Yes	No (FDSOI)	No (FDSOI)	Yes	Yes
Support Bit-Interleaving?	4 (fixed)	No	Yes	Yes	4 (fixed)	No	No	Yes	Yes	Yes	Yes	No	Yes	2 (fixed)	Yes
SRAM Capacity	128Kb	16Kb	16Kb	512Kb	256Kb	8Kb	8.91Mb	2Mb	1Mb	32Kb	32Kb	64kb	64Mb	128Kb	64Kb
Cell Area (μm^2)	2.16 ¹	12.44 ¹	0.8	0.43	1.06 ¹	1.06 ¹	0.82 ¹	0.12	0.12	N/A	0.12	1.01 ¹	0.11	N/A	0.064
Cell Area Increase vs Foundry HD ²	3.79×	21.82×	1.86×	1×	2.47×	4.41×	3.42×	1×	1×	N/A	1×	8.42×	1×	N/A	1.125×
Leakage Power / Bit	32.4fW (@1.2V)	3.17fW (@0.3V)	97.7pW (@0.2V)	0.38pW	55.2fW (@0.9V)	6.25pW (@0.4V)	1.03pW (@0.58V)	3.5pW (@0.7V)	4pW (@0.6V)	12.6pW (@0.8V)	109pW (@0.33V)	30pW (@0.4V)	4pW (@0.8V)	33.6pW (@0.6V)	5.34pW (@0.5V)
Access Time	7ns (@1.2V)	277.78μs (3.6KHz@0.4V)	250ns (4MHz@0.25V)	100ns (10MHz)	56.82ns (17.6MHz@0.9V)	1μs (@0.4V)	143ns (@0.7V)	N/A	N/A	0.42ns (@1V)	N/A	12.5ns (@0.4V)	N/A	N/A	3.94ns (@0.5V)
Access Energy / Bit	391fJ (@1.2V)	1.33fJ (@0.3V)	5.2fJ (@0.25V)	2.5pJ	N/A	11.3fJ (@0.4V)	67fJ (@0.6V)	N/A	N/A	N/A	93fJ (@0.6V)	28fJ (@0.4V)	N/A	67.2fJ (@0.6V)	24.6fJ (@0.5V)

¹logic design rule

²HD cell areas for each process node: 0.57 μm^2 (65-nm), 0.43 μm^2 (55-nm), 0.24 μm^2 (40-nm), 0.12 μm^2 (28-nm), 0.081 μm^2 (20-nm), and 0.064 μm^2 (14-nm)

³Including dummy rows

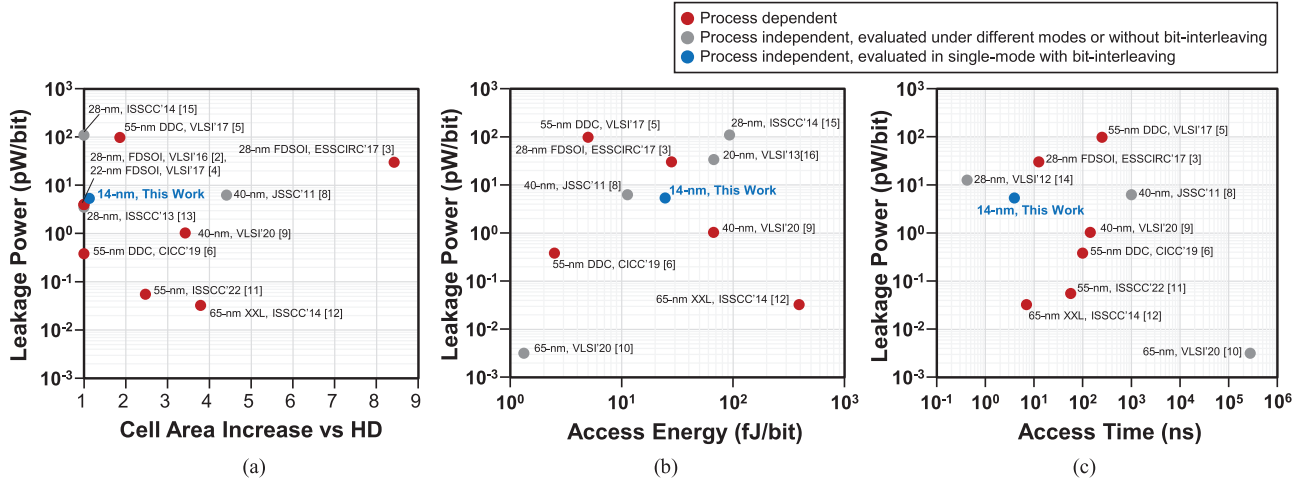


Fig. 20. Landscape of leakage power versus. (a) Cell area increase. (b) Access energy. (c) Access time for low-leakage and low-power SRAMs.

due to limited V_{TH} modulation and requires an extra voltage source for body biasing.

On the other hand, the proposed FVS SRAM does not rely on leakage-specialized processes or techniques like reverse body biasing. In addition, the proposed FVS SRAM is the only design that supports bit-interleaving, operates in single mode, and can be implemented independently of the process nodes. The single-mode supply operation of the proposed FVS SRAM offers substantial energy and latency benefits for IoT systems with frequent transitions between normal and standby modes, although it may require minor adjustments to the conventional voltage control system. Fig. 20(a)–(c) shows the landscape of leakage power versus cell area, access energy, and access time, respectively, for low-leakage and low-power SRAMs. These metrics are particularly critical for modern IoT applications, where the integration of large SRAM capacity significantly affects overall system efficiency. The cell area is closely related to chip cost, access energy directly influences battery life, and access time determines system responsiveness.

Therefore, optimizing these metrics is essential for achieving energy-efficient and reliable performance in IoT devices. The proposed FVS SRAM provides a well-balanced design space, making it highly suitable for energy-constrained IoT systems.

V. CONCLUSION

This work proposed an effective solution for implementing the VS method in SRAM that reduces leakage without the need for an inefficient LDO. The proposed FVS SRAM achieves high area efficiency by using compact HD and pMOS PG cells with improving V_{MIN} by the ICV assist technique and reducing access energy by CS-PCH. The proposed FVS SRAM consumes 5.34-pW/bit leakage power and 24.6-fJ/bit access energy at $V_{MIN} = 0.5$ V and reduces leakage power by 29%–59% and access energy by 10%–21% across $VDD = 0.6$ – 0.8 V compared to conventional non-stacked SRAM. The proposed FVS SRAM is easily adaptable to any process node with the addition of the pMOS PG cell, CVC, and transmission gate-based PC. The single-mode operation is

particularly advantageous for IoT applications with frequent transitions between active and sleep modes. The proposed FVS SRAM also supports bit-interleaving, which improves area efficiency and provides soft error protection, making it a highly attractive option for modern IoT systems.

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